

Prompt Engineering for Data

DS205.3 – Data Science in Python

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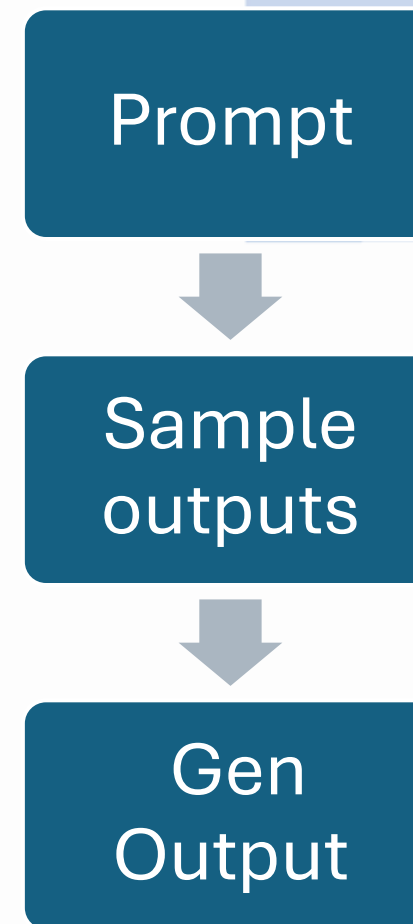
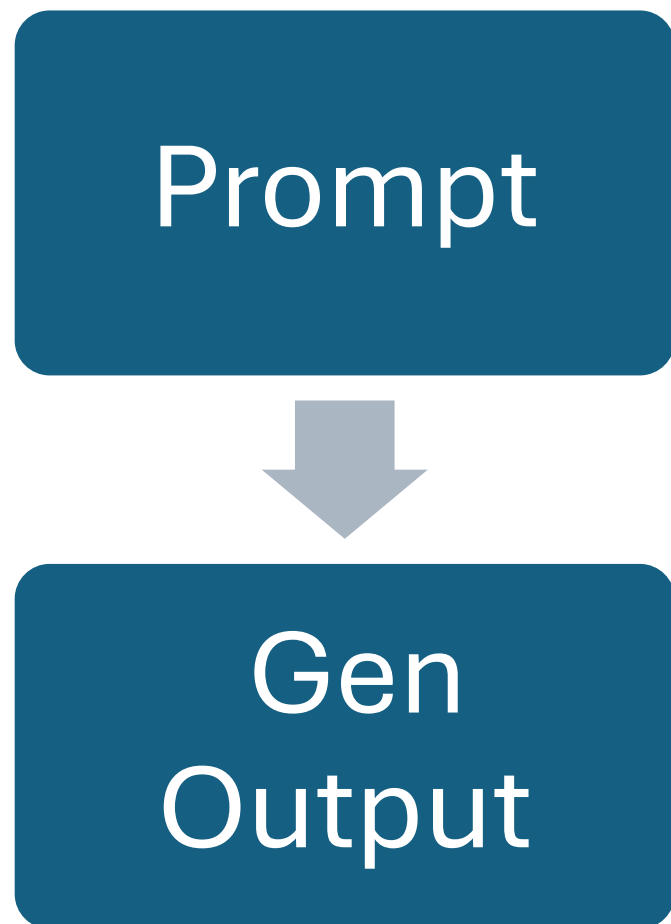
Foundations of Prompt Engineering

- The systematic process of designing and refining inputs to guide an LLM to produce accurate, consistent, and structured outputs.
- Prompt engineering allows the domain to be expanded from non-deterministic "chatting" to deterministic "data extraction."
 - System Prompts: Setting the "Behavioral Constraint."
 - Few-Shot Prompting: Providing a few input-output pairs to minimize variance in output.
 - Chain-of-Thought: Forcing the model to perform intermediate logical reasoning.

Few-Shot Prompting

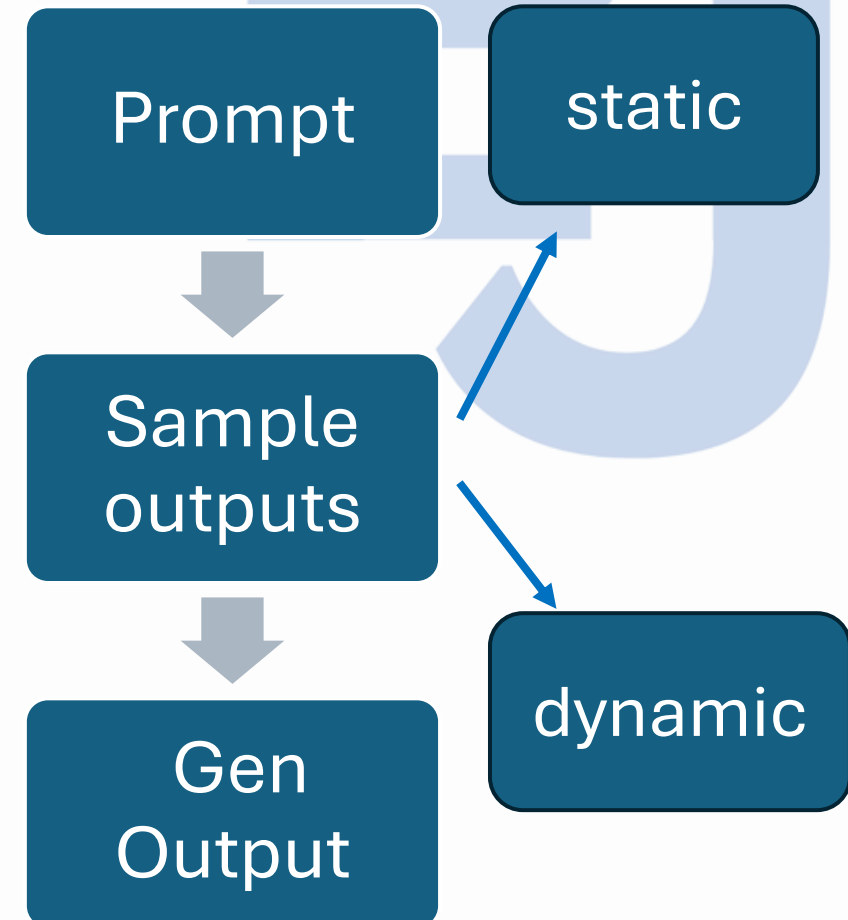
- Few-shot prompting is an AI strategy in which a model's answer is guided by a limited number of examples within the prompt.
- The model can better comprehend the problem and produce more accurate and pertinent results without further training by learning from these examples.
- It reduces the cognitive load on the model and aligns the output format with your system's requirements.

Zero-shot prompting vs Few-Shot Prompting



Few-Shot Prompting mechanism

- The model is guided by examples in few-shot prompting, which can be either predetermined or dynamically fetched depending on the query.
- Static Examples: These are handwritten examples that are directly included in the prompt to provide a clear demonstration of the task.
- Dynamic Examples: To ensure that the examples closely match the user query context, they are retrieved from a vector store utilizing semantic similarity.



Exercise

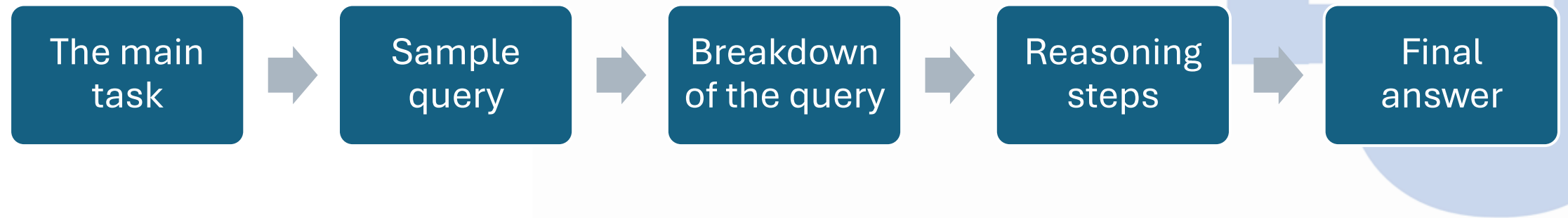
- Generate item descriptions for a teddy bear with zero shot and few shot prompting mechanisms.



Chain-of-Thought Prompting

- Instead of $P(\text{Output} \mid \text{Input})$, we use $P(\text{Output} \mid \text{Reasoning}, \text{Input})$.
- By giving an explanation, the model identifies dependencies in the data that a direct answer might miss.
- Essential for EDA and statistical inference tasks where the LLM must first read the dataframe properties before making a calculation.

Chain-of-Thought Prompting



Applications of the above

- Content creation: Produces text by imitating the styles of examples.
- Video action recognition: Uses little information to classify actions.
- Answering questions: Uses sample Q&A pairs to increase accuracy.
- Translation: Uses a few example pairs to convert text.
- Chatbots: Gain better responses by learning from sample conversations.
- Sentiment analysis uses a small amount of labeled data to classify text.
- NER: Uses a small number of annotated examples to identify entities.
- Code tasks: Helps with debugging and code generation.

Conclusion

- Data pipelines need deterministic output, whereas LLMs are stochastic. The link between them is prompt engineering.
- The ability of a model to learn from the prompt's examples without changing its internal weights is known as "in-context learning".
- Token Budget: The LLM's context limit. Each character or word in your Few-Shot examples uses up this "budget."